

Distinguishing AI-Generated Text from Human Writing:

A Comparative Study of Decision Tree Stylometry and BERT-MLP Classification

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Introduction to Artificial Intelligence • Dr. Syed Ali Raza • Spring 2026

Abstract

The proliferation of large language models (LLMs) such as GPT-4, LLaMA, and Gemini has made it increasingly difficult to distinguish AI-generated text from human writing, raising urgent concerns around academic integrity, misinformation, and content authenticity. This paper presents a systematic comparative study of two AI detection approaches applied to the MAGE benchmark dataset: (1) a Decision Tree (DT) classifier implemented from scratch and operating on 16 handcrafted stylometric features, and (2) a Multi-Layer Perceptron (MLP) operating on 80-dimensional BERT embeddings compressed via Principal Component Analysis. Both models were trained on a balanced corpus of 170,000 samples spanning seven writing domains and approximately 29 generative models. Through iterative dataset refinement across three training phases, a 40configuration parameter sweep, 5-fold stratified cross-validation, and structured inference testing on 11 real-world samples, we find that the BERT-MLP achieves 91.78% test accuracy and ROCAUC of 0.9751, substantially outperforming the Decision Tree which reaches 79.01% accuracy and AUC of 0.8882. The Decision Tree nonetheless provides interpretable feature importances, with hapax ratio (0.4762) and period count (0.1420) emerging as the most discriminative stylometric signals. Both models exhibit a systematic false-positive bias on formal academic writing — precisely the domain most critical for the intended deployment context of academic integrity verification.

Keywords: AI text detection, decision tree, stylometric features, BERT embeddings, MLP, MAGE dataset, academic integrity, natural language processing, binary classification

1. Introduction

The rapid advancement of large language models (LLMs) such as GPT-4, LLaMA, Claude, and Gemini has enabled the generation of fluent, coherent, and contextually appropriate text at an unprecedented scale. While these capabilities unlock genuine productivity gains, they simultaneously introduce acute challenges:

students may submit AI-written work as their own; news outlets may be flooded with synthetic articles; and social platforms may host AI-crafted influence operations indistinguishable from genuine human discourse.

This paper approaches the problem as a binary classification task: given a text passage, determine whether it was written by a human (label 0) or generated by an AI system (label 1). We deliberately compare two methodologically distinct approaches that operate at fundamentally different levels of abstraction.

The first pipeline is a Decision Tree (DT) classifier implemented entirely from scratch in Python, operating on 16 hand-engineered stylometric features that quantify writing style independently of topic content. These features — ranging from hapax ratio and sentence burstiness to transition density and contraction rate — encode the statistical hypothesis that humans and AI systems produce text with measurably different surface-level patterns, even when the content is semantically similar.

The second pipeline is a Multi-Layer Perceptron (MLP) operating on deep contextual embeddings produced by BERT (Bidirectional Encoder Representations from Transformers), a pre-trained transformer model. Unlike the DT, the MLP operates on 80-dimensional semantic representations obtained by compressing BERT's 768-dimensional output via Principal Component Analysis, capturing meaning, coherence, and discourse structure at a level far beyond surface stylometry.

A central motivation for studying both approaches is their different failure modes. The stylometric Decision Tree is lightweight, interpretable, and computationally inexpensive but may miss subtle semantic signals that distinguish sophisticated AI text from human writing. The BERT-MLP captures richer semantic representations but is computationally heavier, less transparent, and vulnerable to out-of-distribution language varieties. Understanding when each approach succeeds or fails provides actionable insight into the nature of AI-generated text itself.

The project also documents an iterative dataset evolution across three training regimes: beginning with a small single-domain corpus (HC3 Reddit QA with 9,000 samples per class), progressing through a larger but domain-narrow dataset (34,000 samples including persuasive essays), and ultimately training on the MAGE benchmark — a 170,000-sample corpus spanning seven domains and approximately 29 generative models. This progression reveals how distributional shortcuts inflate evaluation metrics and how genuine generalisation performance is substantially lower than initially measured.

Core Research Questions

1. Can hand-crafted stylometric features encoded in a Decision Tree distinguish AI-generated text from human writing at a level useful for academic integrity applications? 2. How much does contextual semantic embedding (BERT) improve classification performance relative to surface-level stylometric analysis? 3. What are the characteristic failure modes of each approach, and what do they reveal about the nature of AI-generated text?

2. Related Work

Research on AI-generated text detection has grown rapidly since the emergence of GPT-2. Three papers most directly inform our methodology and experimental design.

2.1 HC3: Human-ChatGPT Comparison Corpus (Guo et al., 2023)

Guo et al. introduced the HC3 dataset and studied stylometric differences between human and ChatGPT responses across Reddit, Wikipedia, and finance domains. Their finding that ChatGPT responses tend to be longer, lexically richer, and structurally more uniform than human counterparts directly motivated our decision to move away from HC3. In our Phase 1 experiments, human answers in HC3 were consistently shorter than AI answers, creating a spurious accuracy inflation via a length confound that allowed near-perfect classification by simple thresholding on text length alone — a scientifically invalid result that we explicitly document and resolve in Phase

3.

2.2 MAGE: Machine-Generated Text Detection in the Wild (Wang et al., 2024)

Wang et al. presented the MAGE benchmark used as our final training and evaluation corpus. MAGE provides 170,000 samples across seven domains — news articles, creative writing, student essays, academic abstracts, social media posts, question-answer pairs, and product reviews — generated by approximately 29 language models including GPT-3.5, GPT-4, LLaMA, Claude, and Gemini. Critically, Wang et al. found that all evaluated methods degrade significantly when tested on domains or models not seen during training — a finding our false-positive experiments corroborate and extend to the specific context of Pakistani academic writing styles.

2.3 Authorship Attribution for Neural Text Generation (Uchendu et al., 2020)

Uchendu et al. demonstrated that features related to lexical diversity, sentence length variance, and punctuation density are statistically significant discriminators between human and machinegenerated text. This work directly informed our feature engineering — particularly the inclusion of hapax ratio, sentence burstiness, and transition density in our 16-feature set. Our results confirm their central finding: hapax ratio (the fraction of words appearing exactly once) is the single most powerful stylometric discriminator, accounting for 47.62% of the Decision Tree's total feature importance.

2.4 BERT: Pre-training of Deep Bidirectional Transformers (Devlin et al., 2019)

Devlin et al.'s foundational work introduced BERT's bidirectional pre-training strategy and demonstrated state-of-the-art performance across 11 NLP benchmarks. Our use of bert-baseuncased as a feature extractor — rather than fine-tuning it end-to-end — is a deliberate design choice motivated by computational constraints. Mean pooling over all non-padding token embeddings (rather than using only the [CLS] token) has been shown in subsequent literature to produce more robust document-level representations for variable-length texts, which is why we adopt this approach in our BERT extraction pipeline.

3. Methodology

3.1 Dataset Evolution Across Three Training Regimes

One of the most instructive aspects of this project is its iterative dataset evolution, in which each training phase was prompted by a diagnosed failure in the previous iteration. This progression from inflated to realistic performance metrics constitutes a research contribution in itself.

3.1.1 Phase 1 — HC3 Reddit QA (9,000 Samples per Class)

The initial training phase used the HC3 Reddit QA subset, providing 9,000 human and 9,000 AI-generated responses. Both the Decision Tree and MLP achieved remarkably high test accuracies of 96% and 99% respectively. Post-training inference analysis revealed the cause: both models had learned a length confound. AI answers in HC3 are systematically 3-4x longer than human QA responses, allowing near-perfect separation by thresholding on word count alone. This result was scientifically invalid — the classifier was not detecting stylistic signals of AI authorship but rather exploiting a domain-specific artifact. Phase 1 models were accordingly discarded.

3.1.2 Phase 2 — HC3 + Persuasive Essays Dataset (34,000 Samples)

Phase 2 expanded the training corpus to 34,000 samples by incorporating a Kaggle persuasive essays dataset. Test accuracy improved to 89% (DT) and 99% (MLP). However, forensic analysis revealed a second data quality issue: approximately 90% of training samples came from a single writing domain — student persuasive essays. This created severe domain overfitting. The models performed well on essay-style text but produced frequent false positives on out-of-distribution text types including news articles, casual conversational messages, and formal academic writing. Phase 2 models were deemed unsuitable for general deployment.

3.1.3 Phase 3 — MAGE Dataset (170,000 Samples) — Final Configuration

The MAGE benchmark provided 85,000 human and 85,000 AI samples across seven distinct writing domains: news articles, creative writing, student essays, academic abstracts, social media posts, question-answer pairs, and product reviews. The balanced and multi-domain corpus forced both models to learn genuinely generalizable signals rather than domain-specific artifacts. Test accuracy settled at 79.01% (DT) and 91.78% (MLP), which represent realistic evaluations of model performance.

Key Methodological Insight

Phase 3 results represent a realistic evaluation of model performance. The accuracy decrease relative to earlier phases — from 96%/99% to 79%/92% — is not model failure. It is the removal of spurious shortcuts that artificially inflated earlier metrics. A classifier that achieves 99% accuracy by learning text length is useless in deployment; a classifier that achieves 79% by learning genuine stylometric signals is genuinely useful.

3.2 Feature Engineering for the Decision Tree

The Decision Tree operates on 16 stylometric features computed directly from raw text. Each feature was selected based on theoretical grounds from prior literature and computational tractability. The features are

designed to capture writing style independently of topic content, so that the classifier's decisions reflect authorship patterns rather than subject matter.

Feature	Definition	Motivation
avg_sent_len	Mean token count per sentence	AI tends toward uniform sentence length
std_sent_len	Std. deviation of sentence lengths	Human writing is more variable in rhythm
avg_word_len	Mean character count per word	AI may prefer longer, more formal vocabulary
std_word_len	Std. deviation of word lengths	Lexical diversity signal
ttr	Type-Token Ratio (unique/total tokens)	Proxy for vocabulary breadth
comma_count	Commas per document	Punctuation rhythm signal
period_count	Periods per document	Sentence boundary frequency (2nd most important)
exclaim_count	Exclamation marks	Emotional expressivity marker
question_count	Question marks	Rhetorical structure indicator
sent_burstiness	Coefficient of variation of sentence lengths	Non-uniform pacing in human text
para_len_variance	Variance in paragraph lengths	Structural layout diversity
hapax_ratio	Fraction of words appearing exactly once	Dominant signal: 47.62% importance
contraction_rate	Rate of contractions (don't, it's)	Informality marker; rare in AI formal text
poly_ratio	Fraction of words with 3+ syllables	Polysyllabic word density
Feature	Definition	Motivation
passive_ratio	Fraction of passive-voice sentences	AI tends toward more passive constructions

transition_density	Rate of discourse transition words	Structural cohesion signal
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Table 1: Complete stylometric feature set used in the Decision Tree classifier

3.3 Decision Tree — From-Scratch Implementation

The Decision Tree was implemented entirely from scratch in Python without use of scikit-learn's tree module, satisfying the course requirement that curriculum-taught methods must be selfimplemented. The implementation covers all components from node structure to the CART split search algorithm.

3.3.1 Node Structure and Tree Representation

Each node in the tree stores the following information: the feature index and threshold value used for the split decision; left and right child pointers; a probability vector of shape (n_classes,) representing the class distribution at that node; the Gini or entropy impurity value at that node; and the count of training samples that reached that node. Leaf nodes are identified by a null feature index and return their stored probability vector for classification, which is then compared against the configurable decision threshold (default 0.70 in production) to produce a binary label.

3.3.2 Impurity Measures

Impurity is computed via either the Gini index or Shannon entropy, controlled by the criterion hyperparameter. For a node with class distribution $p = (p_0, p_1)$, the two measures are defined as follows. Both measures equal zero for pure nodes and are maximized when classes are equally represented.

$$\text{Gini}(p) = 1 - \sum p_i^2 \quad \text{Entropy}(p) = -\sum p_i \log_2 p_i$$

3.3.3 Greedy Split Search — CART Algorithm

At each node, an exhaustive search is performed over all 16 features and all candidate thresholds (midpoints between consecutive unique sorted values in the training data). The information gain for each (feature, threshold) pair is computed as the reduction in weighted impurity achieved by the proposed split. The split maximizing information gain is selected. If no split yields positive gain, the node is designated a leaf. The full CART information gain formula is:

$$\Delta = \text{Impurity}(\text{parent}) - (nL/n) \times \text{Impurity}(\text{left}) - (nR/n) \times \text{Impurity}(\text{right})$$

3.3.4 Stopping Criteria

Three conditions prevent overfitting during tree construction: (i) node purity — if all samples at a node belong to one class, the node becomes a leaf immediately; (ii) maximum depth — controlled by the max_depth hyperparameter, swept over {5, 10, 15, 20, None}; and (iii) minimum sample count — nodes with fewer than min_samples_split training samples become leaves, swept over {2, 10, 50, 100}. The combination of maximum depth and minimum sample count jointly controls the bias-variance tradeoff.

3.3.5 Classification Threshold

By default, inference classifies a sample as AI-generated if $P(\text{AI}) \geq 0.5$. However, in an academic integrity context, a false positive — labelling genuine human text as AI-generated — carries substantially more

consequence than a false negative (missing a case of AI use). A false positive constitutes a potentially unjust accusation of academic dishonesty against a student. We therefore raise the inference threshold to 0.70, conservatively reducing the false positive rate at the cost of a modest increase in the false negative rate. This is a principled ethical design choice, not an ad hoc calibration.

3.4 BERT-MLP Pipeline

The second pipeline operates at the level of semantic meaning rather than surface style. Raw text is first encoded using the bert-base-uncased pre-trained model to produce 768-dimensional contextual embeddings. Unlike approaches that use only the [CLS] token representation, we apply mean pooling over all non-padding token embeddings, which produces more robust document-level representations for variable-length texts.

Because 768 dimensions is high-dimensional relative to the number of training samples and the capacity of a simple MLP classifier, we apply Principal Component Analysis to reduce the embedding space to 80 dimensions while retaining the most informative variance structure. The PCA transformation is fitted on the training set only and stored as bert_pca.pkl, ensuring that the same transformation is applied at inference time without data leakage.

The MLP architecture consists of four layers with progressive dimension reduction: Input(80) → Dense(256, ReLU) → Dense(128, ReLU) → Dense(64, ReLU) → Output(2, Softmax). The model is trained with the Adam optimizer (learning rate $\eta = 0.001$), L2 regularization ($\lambda = 10^{-4}$), batch size of 512, and early stopping with patience of 10 epochs and tolerance of 10^{-4} . Training converged after 22 epochs in 29.4 seconds.

3.5 Experimental Protocol

Both models share a stratified 80/20 train/test split with a fixed random seed to ensure reproducibility. The Decision Tree undergoes a full 40-configuration parameter sweep (5 depths $\times$ 4 min_split thresholds $\times$ 2 impurity criteria), with the best configuration selected by held-out ROC-AUC, followed by 5-fold stratified cross-validation. The MLP uses a 70/15/15 train/validation/test split (118,995 / 25,505 / 25,500 samples respectively) with 5-fold crossvalidation on the combined train and validation pool.

After training, both models were tested on 11 real-world text samples not drawn from the MAGE training distribution, covering a range of registers, topics, and linguistic varieties including casual personal writing, formal academic essays, AI literary fiction, news-style text, Roman Urdu, and adversarially modified AI text with intentional typographic errors.

4. Results and Discussion

4.1 Decision Tree — Parameter Sweep Results

The full 40-configuration parameter sweep was conducted on the MAGE 170,000-sample test set. Representative results are shown in Table 2. The winning configuration — depth=15, min_samples_split=100, criterion=Gini — achieved AUC=0.8882 and F1=0.7760. The high min_samples_split value of 100 acts as implicit regularization, preventing nodes from memorizing small sub-populations of training data. Notably, deeper configurations (depth=20 and depth=None) achieved lower AUC despite higher recall, confirming that deeper trees overfit in the stylometric feature space — the additional splits learn noise rather than signal.

Configuration	Acc	Prec	Rec	F1	AUC	Time(s)
depth=5, minsplitt=2, gini	0.6786	0.7532	0.5314	0.6231	0.7983	1.04
depth=10, minsplitt=2, gini	0.7722	0.8076	0.7147	0.7583	0.8666	2.06
depth=15, minsplitt=10, gini	0.7973	0.8193	0.7628	0.7900	0.8696	2.94
depth=15, minsplitt=50, gini	0.7937	0.8279	0.7415	0.7823	0.8839	2.93
depth=15, minsplitt=100, gini ★	0.7901	0.8320	0.7270	0.7760	0.8882	2.84
depth=15, minsplitt=100, entropy	0.7850	0.8326	0.7134	0.7684	0.8863	3.11
depth=20, minsplitt=2, gini	0.7939	0.7943	0.7932	0.7938	0.8089	3.56
depth=None, minsplitt=2, gini	0.7790	0.7832	0.7715	0.7773	0.7790	4.11

Table 2: Selected configurations from the 40-configuration DT sweep on MAGE 170k test set (★ = best by AUC)

4.2 Decision Tree — Cross-Validation Results

Following the parameter sweep, 5-fold stratified cross-validation was performed on the best configuration. The results confirm stable generalization with extremely low variance across folds.

Metric	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	CV Mean ± Std	Test Set
Accuracy	0.8092	0.8115	0.8119	0.8103	0.8101	0.8106 ± 0.0019	0.7901

Precision	0.8641	0.8653	0.8659	0.8648	0.8645	0.8649 ± 0.0007	0.8320
Recall	0.7292	0.7319	0.7321	0.7305	0.7302	0.7308 ± 0.0012	0.7270
Metric	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	CV Mean ± Std	Test Set
F1 Score	0.7908	0.7932	0.7935	0.7914	0.7911	0.7920 ± 0.0012	0.7760
ROC-AUC	0.8809	0.8831	0.8843	0.8820	0.8822	0.8825 ± 0.0020	0.8882

Table 3: 5-fold stratified cross-validation results vs. held-out test set (threshold = 0.70)

Cross-Validation Finding

The very low standard deviation across folds (± 0.0007 to ± 0.0020) confirms stable generalisation with no signs of overfitting. The ROC-AUC is marginally higher on the independent test set (0.8882) than the CV mean (0.8825), a threshold-independent result confirming that the final model trained on the full training set is not overfit to any particular fold.

4.3 Feature Importances — Decision Tree

The normalized feature importance scores extracted from the final Decision Tree model provide interpretable insight into what stylometric signals most reliably differentiate AI from human text.

Feature	Importance	Interpretation
hapax_ratio	0.4762	Dominant signal: AI reuses standardized vocabulary; humans introduce more unique words
period_count	0.1420	AI tends to write shorter, more complete sentences with higher sentence boundary density
sent_burstiness	0.0685	Human writing has more irregular sentence length patterns
ttr	0.0663	Type-token ratio: correlated with but distinct from hapax ratio
avg_sent_len	0.0532	AI sentences tend to be slightly longer on average
std_sent_len	0.0508	Lower in AI text (more uniform sentence lengths)

transition_density	0.0356	AI text uses discourse markers more systematically
avg_word_len	0.0319	AI tends toward slightly longer, more formal vocabulary
passive_ratio	0.0214	AI more frequently employs passive voice constructions
poly_ratio	0.0198	AI uses more complex polysyllabic vocabulary
contraction_rate	0.0089	AI rarely uses contractions in formal text
para_len_variance	0.0062	Near-zero for many single-paragraph inputs
comma_count	0.0050	Low discriminative value; largely stylistic variation
exclaim_count	0.0014	Very rare in both classes in formal text
question_count	0.0012	Marginal discriminator
Feature	Importance	Interpretation
std_word_len	0.0008	Near-zero importance; correlates strongly with avg_word_len

Table 4: Normalised feature importances from the final DT model (MAGE 170k, depth=15, minsplit=100, Gini)

4.4 MLP Training Convergence

The MLP trained on BERT embeddings demonstrated rapid convergence with monotonically decreasing training loss. Key convergence statistics are summarized below. Training was performed on a 70/15/15 stratified split of the 170,000-sample MAGE corpus.

Epoch	Training Loss	Validation Accuracy
1	0.35440381	0.895714
2	0.22077458	0.905126
5	0.15789032	0.914118
8	0.12117975	0.919916
11	0.09317573	0.920336 (peak)

15	0.06398047	0.915378
20	0.04019600	0.912689
22	0.03630249	0.911849 (stopped)

Table 5: MLP training convergence log — early stopping triggered after epoch 22 (patience=10, tolerance=10⁻⁴)

4.5 MLP — Final Evaluation Results

The MLP achieved strong performance across all three evaluation sets. The confusion matrix from the test set reveals the following absolute counts: True Negatives = 11,578 (human correctly identified), False Positives = 1,172 (human misclassified as AI), False Negatives = 923 (AI misclassified as human), True Positives = 11,827 (AI correctly identified).

Metric	Train Set	Validation Set	Test Set
Accuracy	0.9685	0.9180	0.9178
Precision	0.9686	0.9181	0.9180
Recall	0.9685	0.9180	0.9178
F1 Score	0.9685	0.9180	0.9178
ROC-AUC	0.9948	0.9757	0.9751
Metric	Train Set	Validation Set	Test Set
MCC	0.9371	0.8361	0.8358
Log-Loss	0.0896	0.2342	0.2392

Table 6: BERT-MLP evaluation metrics across train, validation, and test sets

4.6 Cross-Phase and Cross-Model Comparison

Table 7 presents the complete cross-phase performance comparison, demonstrating the effect of dataset evolution on measured accuracy and the superior performance of the BERT-MLP pipeline across all metrics.

Metric	DT Phase 1 (HC3 9k)	DT Phase 2 (34k Essays)	DT Phase 3 (MAGE 170k)	MLP Phase 3 (MAGE 170k)	MLP Gain vs. DT
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Accuracy	0.9600	0.8915	0.7901	0.9178	+0.1277
Precision	0.9623	0.8981	0.8320	0.9180	+0.0860
Recall	0.9578	0.8810	0.7270	0.9178	+0.1908
F1 Score	0.9600	0.8895	0.7760	0.9178	+0.1418
ROC-AUC	0.9910	0.9380	0.8882	0.9751	+0.0869

Table 7: Cross-phase and cross-model performance comparison on held-out test sets

Key Finding

The accuracy degradation from Phase 1/2 to Phase 3 is not model failure but the removal of distributional shortcuts. Phase 1 models exploited a length confound; Phase 2 models overfit to a single essay-writing domain. Phase 3 numbers represent a realistic evaluation on a genuinely diverse, multi-domain corpus. The BERT-MLP achieves 91.78% vs. the DT's 79.01% — a 12.77 percentage-point gap representing the information advantage of semantic embedding over surface stylometry.

5. Inference Testing and Failure Case Analysis

To assess real-world performance beyond the MAGE test set, both models were tested on 11 text samples not drawn from the training distribution. These samples were selected to stress-test specific failure hypotheses identified during development.

Text Description	True Label	DT Prediction	MLP Prediction	MLP P(AI)	Analysis
Casual personal (Siraiki travel fantasy)	Human	Human ✓	Human ✓	0.020	Both correct; high hapax ratio triggers correct leaf
AI literary scene (evening breeze)	AI	AI ✓	AI ✓	1.000	Clean AI signal; strong stylometric + semantic signal
Personal hostel text (Karachi heat)	Human	AI X	Human ✓	0.031	DT false positive; short avg word length lowers hapax signal
AI educational paragraph (definition)	AI	AI ✓	AI ✓	0.997	Both correct; strong stylometric and semantic signal

Roman Urdu casual (Quetta hostel)	Human	Human ✓	AI ✗	0.999	MLP false positive: OOD Roman Urdu not in BERT training
Human student essay (plastic bags)	Human	AI ✗	Human ✓	0.000	DT false positive; formal essay register close to AI domain
Archaic warning message	Human	Human ✓	AI ✗	0.622	MLP borderline false positive; unusual register confuses BERT
Human opinion piece (skills)	Human	Human ✓	Human ✓	0.335	Both correct; longer word count stabilises DT features
Human argument (endangered animals)	Human	AI ✗	Human ✓	0.001	DT false positive; wellstructured argument triggers AI class
Human IELTS-style essay (arts education)	Human	Human ✓	Human ✓	0.065	Both correct; standard IELTS register within distribution
AI sci-fiction (Aethelgard, with typos)	AI	Human ✗	AI ✓	1.000	DT false negative; intentional typos reduce hapax signal

Table 8: Structured inference testing on 11 real-world samples not drawn from the MAGE training set

5.1 Decision Tree Failure Mode — The Formal Register Problem

The Decision Tree produced false positives on three formal human texts (cases 3, 6, and 9 from Table 8). All three shared a characteristic stylometric profile: low hapax ratio (0.25-0.42), long average sentence length (17-22 tokens), elevated passive ratio, and high transition density. This profile overlaps substantially with MAGE's AI-generated training samples, which include formal academic abstracts and structured argumentative text.

The root cause is a domain gap in the MAGE human corpus: MAGE does not include sufficient formal academic or structured argument-style human writing in its human component. The model has therefore learned that formal writing implies AI authorship — a valid signal for the training distribution but a systematic error for the deployment context (academic integrity verification, where students write formal essays).

The sci-fi false negative (case 11 — the Aethelgard text) reveals a complementary adversarial vulnerability: the AI text contained intentional typographic errors and dialogue fragments with short, irregular sentences, which reduced the burstiness and hapax signals that normally discriminate AI text. This demonstrates that

stylographic detection can be systematically evaded by adversarially introducing human-like surface patterns — a significant practical limitation.

5.2 MLP Failure Mode — Out-of-Distribution Language

The MLP produced a false positive on Roman Urdu text (case 5) with $P(AI) = 0.999$. This is an expected and principled failure: MAGE contains no Roman Urdu (transliterated Urdu written in Latin script), and BERT's pre-training corpus does not include this language variety in meaningful quantity. The resulting embedding is essentially random in the learned feature space, and the MLP maps it to the AI class by default — a data coverage gap, not an algorithmic failure.

The archaic warning message (case 7) produced a borderline false positive with $P(AI) = 0.622$. The unusual register — formal, imperative, and archaic — is structurally similar to instructional AI text that BERT has been trained on extensively, causing embedding overlap with the AI class.

5.3 Comparative Error Summary

Model	False Positives	False Negatives	Primary Failure Driver
Decision Tree	3 of 11	1 of 11	Formal human writing misclassified as AI (register overlap)
BERT-MLP	2 of 11	0 of 11	Out-of-distribution language varieties (Roman Urdu, archaic)

Table 9: Error breakdown on 11 real-world inference samples

The DT's false-positive pattern disproportionately affects formal argument-style human writing — precisely the text type most likely to be submitted in academic settings. The MLP's errors are driven by out-of-distribution language varieties, which are arguably edge cases in a typical academic environment. For the intended deployment use case of academic integrity verification at an English-medium institution, the MLP's failure mode is substantially less consequential than the DT's.

6. Discussion

6.1 Why BERT Embeddings Substantially Outperform Stylometry

The 12.77 percentage-point accuracy gap between the BERT-MLP and the Decision Tree reflects a fundamental difference in the information available to each model. Stylometric features capture surface-level writing patterns — word frequency distributions, sentence length variance, punctuation density. These patterns are real and discriminative, as evidenced by the DT's abovechance performance. However, they do not capture the semantic coherence, discourse structure, and logical organization that are characteristic of LLM-generated text at a deeper level.

BERT embeddings encode contextual meaning: the same word in different contexts produces different embedding vectors, and the mean-pooled document embedding captures the overall semantic texture of the text. LLM-generated text has characteristic semantic textures — it tends to be logically coherent, well-organized, and topically consistent in ways that many human writing styles are not. This semantic consistency is what the MLP learns to detect, and it is information that is simply unavailable to a surface-level stylometric analysis.

6.2 The Interpretability-Performance Tradeoff

While the MLP substantially outperforms the DT, the DT provides something the MLP cannot: interpretable feature importances and a traceable decision path. When the DT classifies a text as AI-generated, it is possible to examine exactly which features drove that classification — for example, 'hapax_ratio=0.28 (below threshold 0.35) AND period_count=18 (above threshold 12) AND sent_burstiness=0.09 (low) → AI classification at leaf node.' This level of interpretability is critical in the academic integrity context, where a student who receives an AI-authorship accusation deserves an explanation of the reasoning, not just a probability score.

An ensemble approach combining the DT and MLP would preserve interpretability (via the DT's feature signals) while achieving the MLP's superior accuracy, and is identified as the primary direction for future work.

6.3 Ethical Considerations and Threshold Design

The choice of inference threshold has direct ethical implications. At the default 0.50 threshold, the DT achieves a false positive rate of 0.2389 — meaning approximately 1 in 4 human-written texts would be incorrectly flagged as AI-generated. This is unacceptably high for an academic integrity context, where a false positive constitutes a potentially life-altering accusation. Raising the threshold to 0.70 reduces the FPR to approximately 0.18, at the cost of increasing the false negative rate from approximately 0.13 to approximately 0.17.

This threshold choice reflects a principled ethical position: in the asymmetric consequence space of academic integrity decisions, false accusation (false positive) is more harmful than missed detection (false negative). The 0.70 threshold encodes this value judgment in a formally justifiable way. Any deployment of either model in a real academic integrity context should include human review of all borderline cases (P(AI) between 0.50 and 0.85) and explicit disclosure to students that the tool is probabilistic, not deterministic.

7. Conclusion

This project presents a systematic comparative study of two AI text detection approaches on the MAGE benchmark: a from-scratch Decision Tree operating on 16 stylometric features, and a BERT-MLP pipeline using semantic embeddings. The study documents a complete experimental lifecycle from naive baseline to realistic evaluation, including explicit identification and correction of distributional shortcuts in Phases 1 and 2.

The five key findings are as follows. First, the BERT-MLP substantially outperforms the DT on all metrics — 91.78% vs. 79.01% test accuracy; 0.9751 vs. 0.8882 ROC-AUC — confirming that deep semantic representations capture discriminative structure unavailable in surface stylometric signals. Second, the DT provides actionable feature importances, with hapax ratio (47.62%) and period count (14.20%) emerging as the most discriminative features and providing theoretical insight into the nature of LLM text generation. Third, the dataset evolution from Phase 1 to Phase 3 demonstrates that early high-accuracy results (96-99%) were artifacts of distributional shortcuts, and that realistic generalisation performance on a diverse corpus is substantially lower. Fourth, both models exhibit a systematic false-positive bias on formal academic human writing — the corpus domain least represented in MAGE's human component and most critical for the intended deployment use case. Fifth, the stylometric DT is vulnerable to adversarial evasion via intentional typographic noise, while the MLP is vulnerable to out-of-distribution language varieties such as Roman Urdu.

The recommended paths forward are: (1) augment training corpora with formal academic human writing from pre-2015 journals, classic essays, and legal documents to reduce the DT's falsepositive rate on formal text; (2) include multilingual and code-mixed corpora to reduce the MLP's vulnerability to Roman Urdu and similar varieties; (3) investigate ensemble approaches that combine stylometric interpretability with semantic classification accuracy; and (4) evaluate adversarial robustness more systematically to understand the extent of evasion risk in deployment.

Final Summary

Decision Tree (MAGE 170k): Accuracy = 79.01%, F1 = 0.7760, AUC = 0.8882 BERT-MLP (MAGE 170k): Accuracy = 91.78%, F1 = 0.9178, AUC = 0.9751 Both models are deployable with appropriate threshold calibration and human oversight. The BERT-MLP is recommended for accuracy-critical applications; the Decision Tree is recommended where interpretability and computational efficiency are priorities. An ensemble of both, combined with corpus augmentation to address the formal register gap, represents the optimal path to a productiongrade academic integrity tool.

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